Smart Water Tank Automation

ESP32 hmanga tanky 3, valve 3 leh pump control — mobile app atangin

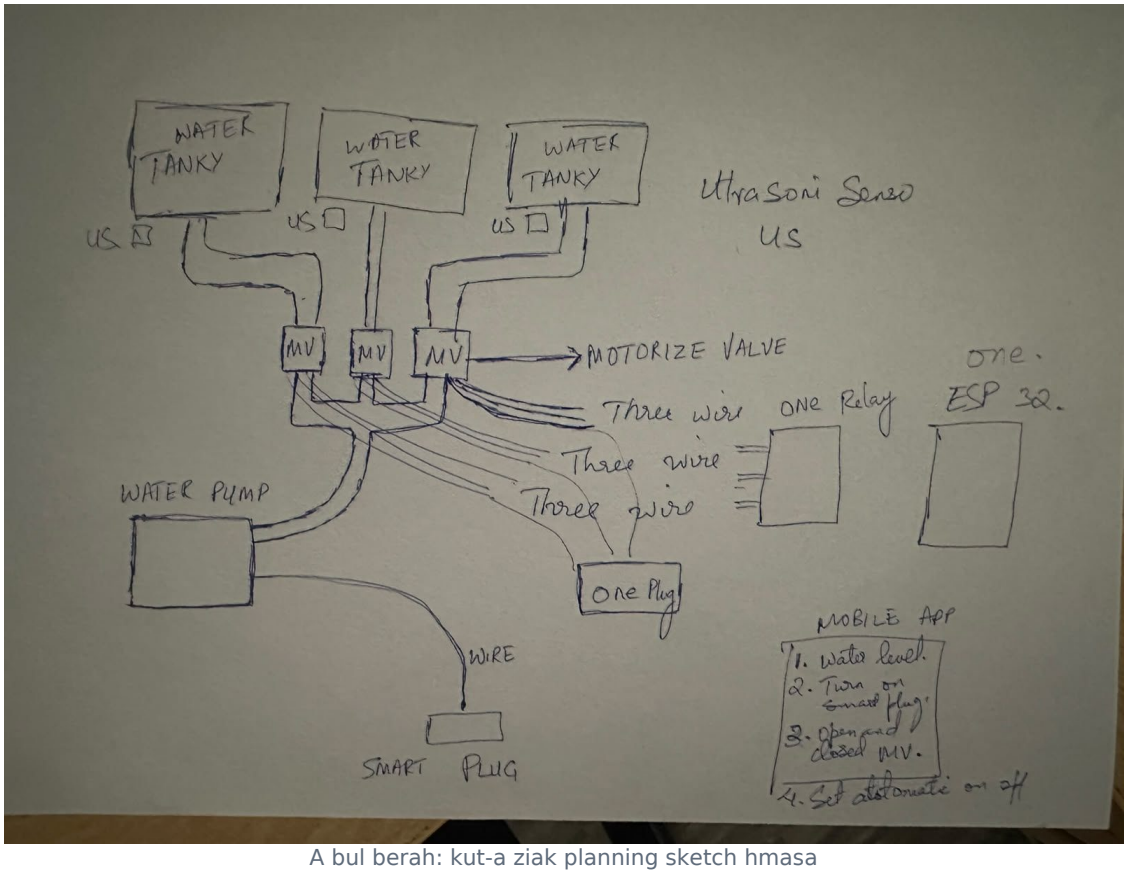


Product installation hmuhtheihna (AI-in a siam, brochure atan)

Thupui	Thil awmte
Hardware	ESP32 pakhat, JSN-SR04T sensor 3, 12V motorized valve 3, relay 4CH, contactor, 12V adapter
Design	Wired (battery awm lo) — main box pump bulah, tank tin CAT6 cable pakhat
App	Water level, pump on/off, valve hawng/khar, auto mode
Man (prototype)	Rs 7,000–8,000 vel
Version / ni	v1.0 · 25 October 2026

1. Idea leh a tum

Tanky 3 (in chung emaw hmun hrang hrangah) chu pump pakhat atangin automatic-in tihkhat a ni ang. Tank tinah ultrasonic sensor pakhat in tui level a teh a, ESP32 pakhat-in valve hawng/khar leh pump on/off a control ang. Mobile app atangin engkim en leh control theih a ni ang, product anga zawrh tura siam a ni.



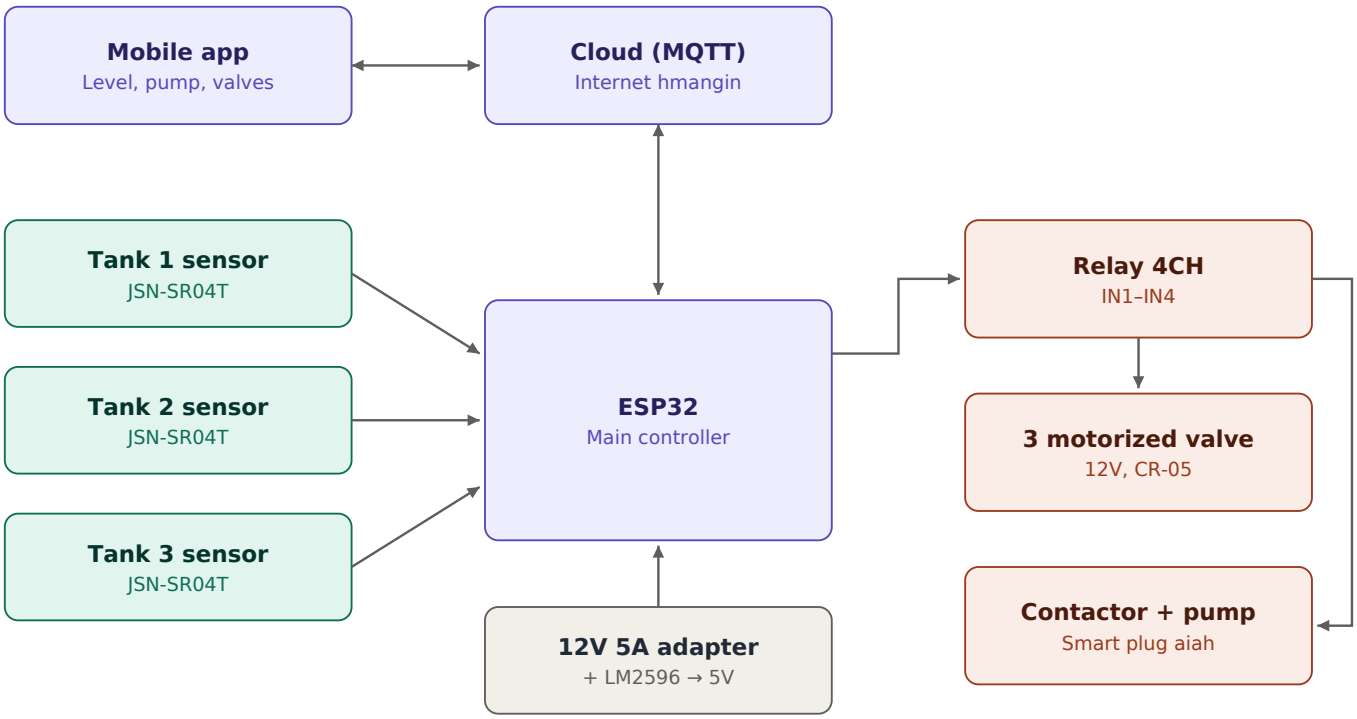
A bul berah: kut-a ziak planning sketch hmasa

Mobile app-in a tih theih tur

- **Water level** — tank 3 te level % in a lang reng
- **Pump on/off** — kut hmangin (manual)
- **Valve hawng/khar** — valve pakhat zel
- **Auto mode** — level hniam chuan a tikhat a, a khat chuan a ti-off ve ve

2. System architecture

ESP32 pakhat hian sensor 3 a chhiar a, relay hmangin valve 3 leh pump contactor a control a, cloud (MQTT) kaltlangin mobile app nen a inbiak.



Thil zawng zawng inzawm dan tawi te

Thutlukna pawimawhte

Thupui	Thutlukna	A chhan
Smart plug	Hmang lo — relay + contactor 25A	Smart plug tam zawk an company cloud-ah an inphuar; pump start current-in 10A plug a tikang thei. Relay + contactor chuan ESP32 leh app pakhat atangin control theih.
Valve	12V motorized ball valve (CR-05, 3-wire)	Zawi muangin a hawng/khar (water hammer awm lo), a hawn hnuah current a ei lo, pressure a ngai lo, kawng a zau.
Sensor	JSN-SR04T (waterproof)	Tank chhung khawhu a tuar; probe leh board a inthen; UART mode a nei.
Power	12V 5A adapter + LM2596 buck (5V)	Valve tan 12V, ESP32/relay/sensor tan 5V.
Battery	Hmang lo (wired)	Sensor tan battery chuan signal wire a paih thei lo; cable pakhatah 5V core a awm tho. A tlawm ber leh buai lo ber.
Cable	Tank tin CAT6 pakhat	Core 8 — 4 hman, a bang spare. Metre khat Rs 15–25 vel.

3. Component thlan dan

Valve: solenoid vs motorized



Brass 12V 2W solenoid — Rs 1,225

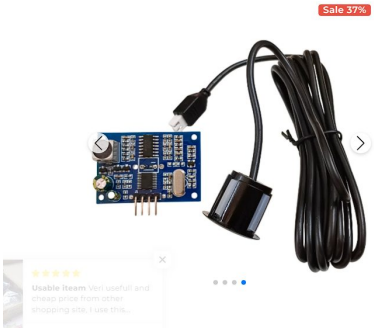


Plastic 12V 1/2" solenoid — Rs 251

	Plastic solenoid	Brass solenoid	Motorized ball valve ✓
Tui luan	Kawng te — pump a hah	A tha zawk	Kawng zau ber
Current	Hawn chhung zawng (lum)	Hawn chhung zawng (lum)	Hawn/khar laiin chauh
Water hammer	Thut khatah khar — a kek thei	A awm	Awm lo (zawi muang)
Ni / ruah	A chhe thuai	A tuar	A tuar
Man (vel)	Rs 251	Rs 1,225	Rs 800–1,500
Tlak dan	Demo/test chauh	A hman theih	Product atan a tha ber

Valve lei dawnin **CR-05 type (wire 3)** lei rawh. CR-02 (wire 2, polarity thlak) chuan valve pakhatah relay 2 a ngai. Valve size hi i pipe size (1" emaw 3/4") nen a inmil tur a ni.

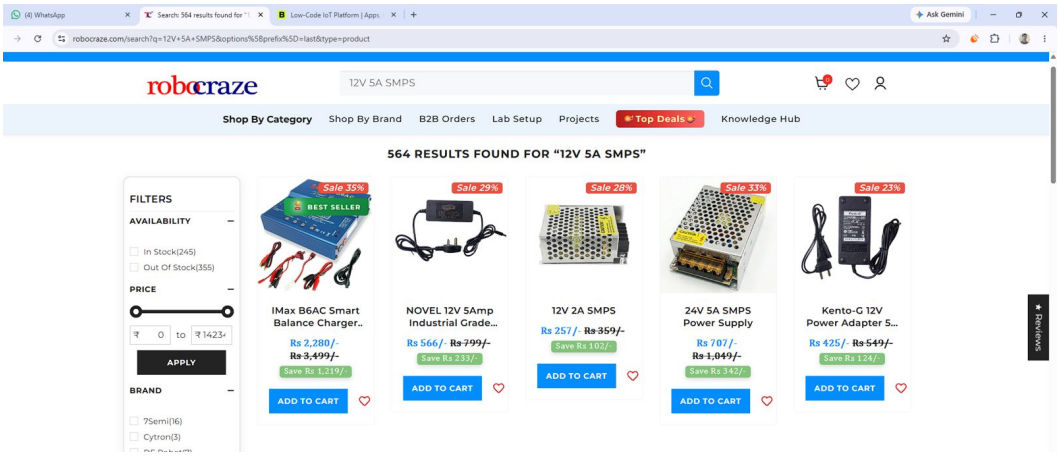
Sensor: JSN-SR04T



	HC-SR04	JSN-SR04T ✓
Waterproof	Aih	Probe a ni
Blind zone	2 cm	20–25 cm
Range	2–400 cm	25–450 cm
Mode	Trig/Echo	Trig/Echo + UART

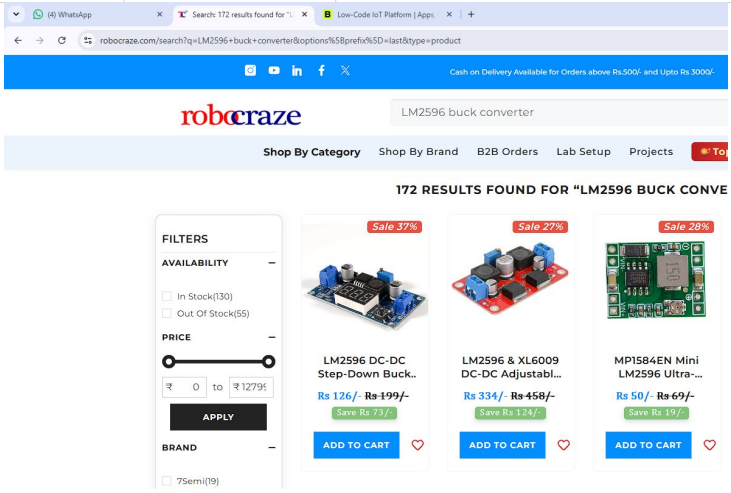
- Probe hi tui khat ber level aiin **cm 30 tal a sang**ah dah (blind zone avangin).
- Echo pin 5V a ni — ESP32-ah luh hmain **1k + 2k voltage divider** dah ngei ngei.
- Board hi waterproof a ni lo — box chhungah dah. Cable metre 10 chuang chuan **UART mode** hmang.

4. Power supply



Robocraze-a 12V SMPS thlan tur

Item	Thlan	A chhan
iMax B6AC	✗	Battery charger a ni
NOVEL 12V 5A — Rs 566	✓	Adapter type, industrial grade
12V 2A SMPS — Rs 257	△	Current a tlem deuh
24V 5A SMPS — Rs 707	✗	Voltage a dik lo
Kento-G 12V 5A — Rs 425	✓ (a tlawm ber)	Adapter type, a him



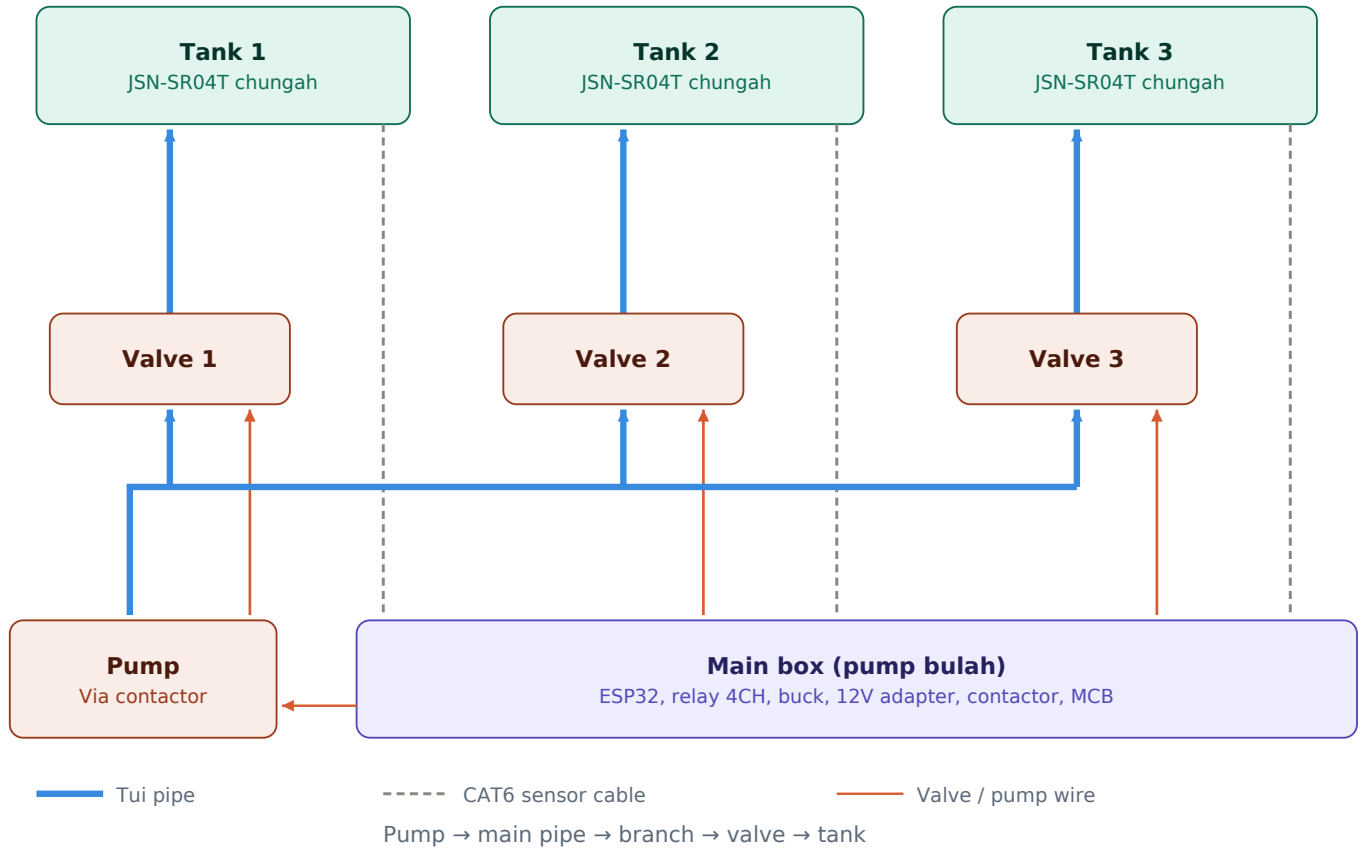
LM2596 buck converter — display nei (Rs 126) hi thlang rawh

Buck converter fimkhur tur: a lei tirh chuan output voltage hriat a ni lo (12V vel pawh a ni thei). ESP32 i zawm hmain a potentiometer her la **5.0V**-ah siam hmasa. Adapter DC jack atan "DC barrel jack female to screw terminal" pakhat lei bawk.

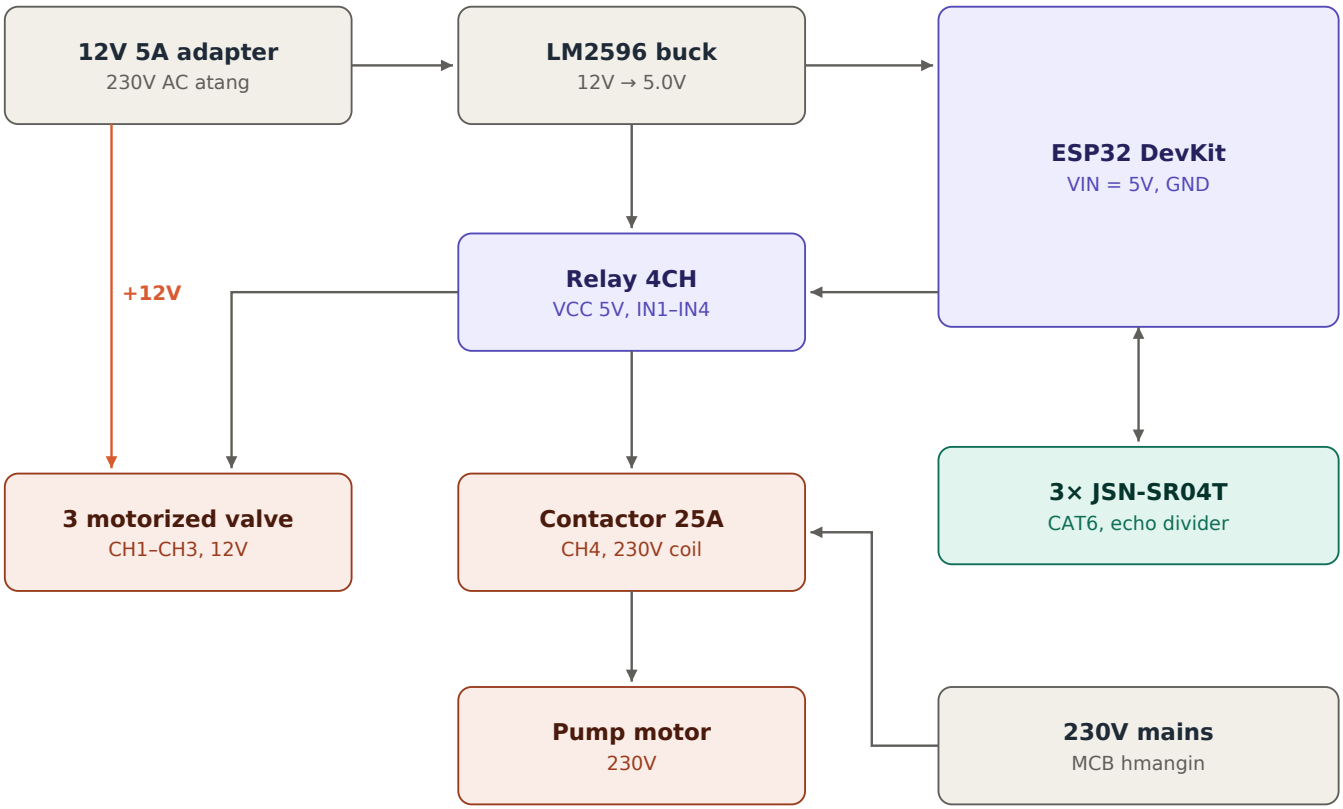
Current mamawh: buck atang (ESP32 + relay + sensor) 0.3A vel (12V-ah) + valve 0.5A vel (motorized) = 1A vel. 5A adapter chuan hlimthla a tam, a lum lo.

5. Site layout

Main box pakhat pump bulah. Tank tin atang CAT6 cable pakhat main box-ah, main box atang valve tin-ah 12V wire.



Main box chhunga wiring



6. Pin leh wire zawm dan

ESP32 pin map

ESP32 pin	Zawm na	Note
VIN	Buck 5V (+)	5.0V a nih ngei a ngai
GND	Buck GND	GND zawng zawng a inzawm vek tur
GPIO 16 / 17	Tank 1 Trig / Echo	Echo divider hmangin
GPIO 18 / 19	Tank 2 Trig / Echo	Echo divider hmangin
GPIO 22 / 23	Tank 3 Trig / Echo	Echo divider hmangin
GPIO 25 / 26 / 27	Relay IN1 / IN2 / IN3	Valve 1 / 2 / 3
GPIO 33	Relay IN4	Pump contactor
GPIO 32	Float switch (duh chuan)	Sump dry-run protection
GPIO 21, 4	Ruak	Nakin belh atan

CAT6 cable (tank tin pakhat)

Wire rong	Sensor lam	Main box lam
Brown	VCC	5V
Brown-white + white dang	GND	GND
Blue	Trig	ESP32 Trig GPIO
Green	Echo	1k → GPIO; GPIO atang 2k → GND

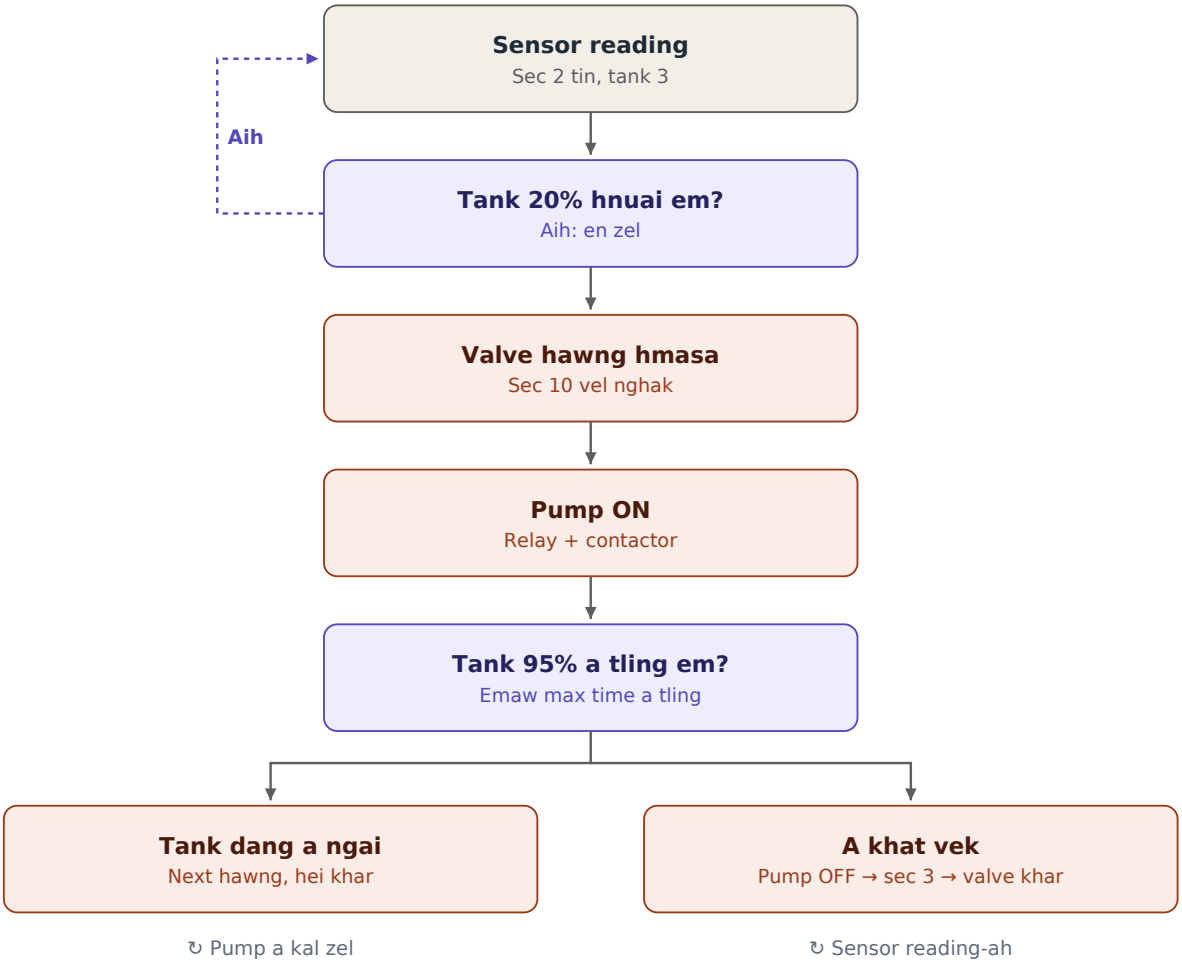
Valve wiring (CR-05, wire 3)

- Wire pakhat **+12V reng**, pakhat **GND**, pakhat **control** (rong hi brand azirin a danglam — valve label en rawh).
- Relay **COM ← +12V**, **NO → valve control**. Relay ON = valve hawng, OFF = khar.

Pump wiring (230V — electrician hmang rawh)

- Relay CH4 **COM ← Phase (L)**, **NO → contactor A1**, **A2 ← Neutral**.
- Pump chu contactor main contact atangin. **MCB** pakhat dah ngei ngei.
- Relay module tam zawk hi **active LOW** a ni (IN = LOW chuan ON) — code-ah ngaihtuah.

7. Automatic mode a kal dan



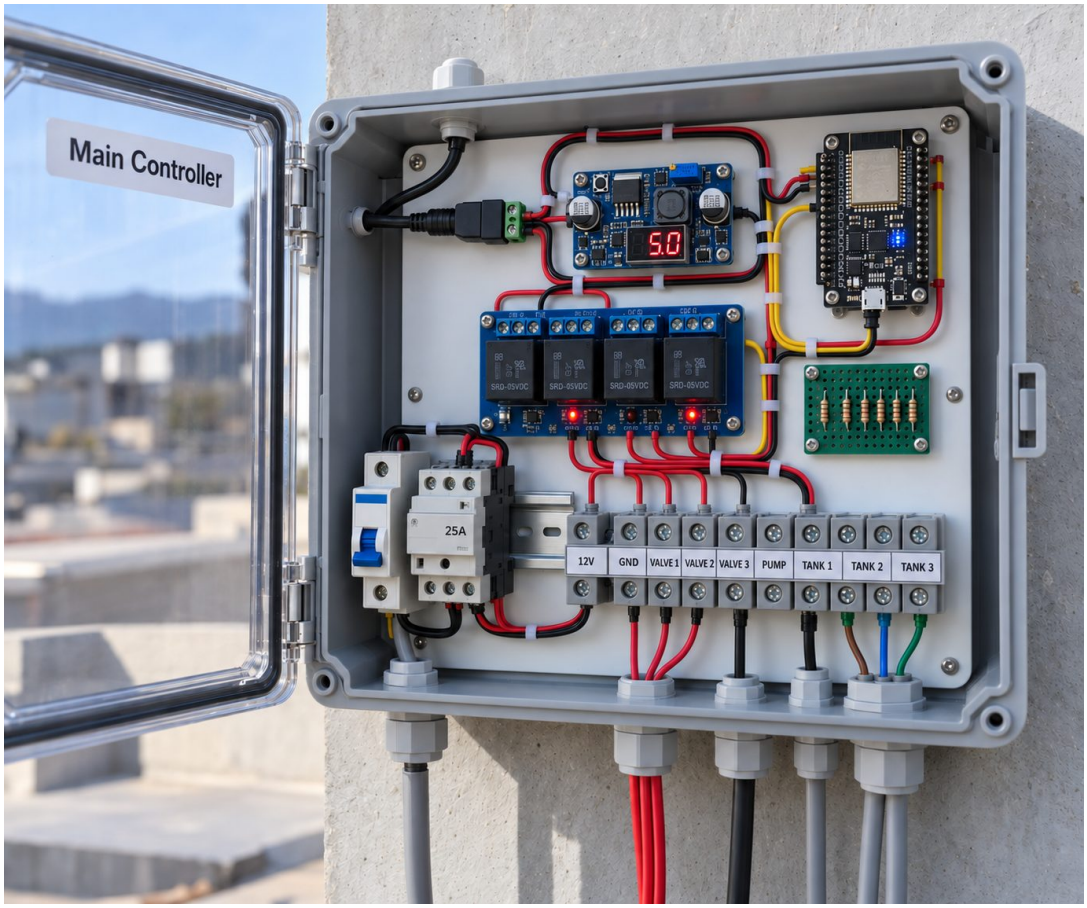
Himna dan (safety rules)

- **Pump a kal laiin valve zawng zawng khar reng a awm ngai lo** — pipe a puak thei, pump a chhe thei.
- Hawn dawnin: **valve hawng hmasa → nghak → pump ON**. Khar dawnin: **pump OFF → nghak → valve khar**.
- Tank pakhat zel tihkhat (valve pakhat chauh hawng) — pump pressure a tha zawk.
- **Max run time** (e.g. min 30) — sensor a chhiat pawhin pump a kal reng lo.
- Sensor reading dik lo (0, emaw range pawn) a awm chuan pump OFF + app-ah alert.
- **Hysteresis:** 20% hnuaiah ON, 95%-ah OFF — on/off zing lutuk loh nan.
- Sump/tui lakna a kang thei chuan float switch belh — dry-run protection.
- App-ah manual override a awm reng; auto mode chu a ti-off theih.

8. Thil leina tur (BOM)

Thil	Zat	Man (vel)	Robocraze search
ESP32 DevKit	1	Rs 400	ESP32 DevKit
Relay module 4CH 5V	1	Rs 150	4 channel relay module 5V
LM2596 buck (display)	1	Rs 126	LM2596 buck converter
12V 5A adapter	1	Rs 425–566	12V 5A SMPS
DC jack to screw terminal	1	Rs 25	DC barrel jack female
JSN-SR04T	3	Rs 1,500	JSN-SR04T
Motorized ball valve 12V CR-05	3	Rs 3,000	motorized ball valve
Contactor 25A, 230V coil	1	Rs 600	Electrical dawr (Schneider/L&T;)
MCB	1	Rs 200	Electrical dawr
CAT6 cable	Inhlat zat	Rs 15–25/m	Electrical/computer dawr
Resistor 1k, 2k (6 zel), perfboard	—	Rs 50	resistor kit
IP65 box, cable gland, terminal block	—	Rs 400	IP65 junction box
A vaiin		**Rs 7,000–8,000**	

Man hi chhut zat a ni — lei hmain dawr-ah check leh rawh.



Main controller box chhung hmuhtheihna (AI-in a siam)

9. App leh software

Option	A tha na	A tha lo na	Hman hun
Blynk IoT	Rang takin a siam theih, app a awm nghal	Branding i duh angin a siam theih lo; plan man	Prototype / test
Nangma PWA + MQTT (HiveMQ Cloud)	I brand, i feature duh ang; customer tam tak	Siam rei deuh	Product zawrh tur

MQTT topic tlangpui

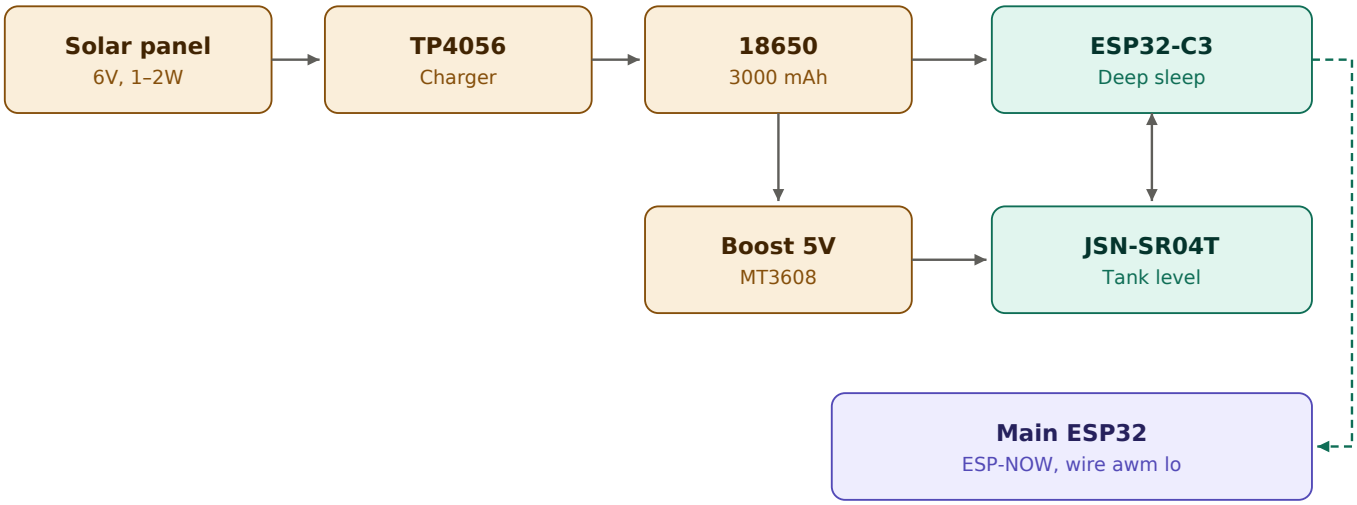
```
tank/<device-id>/level/1      → 75        (ESP32 → app, %)
tank/<device-id>/level/2      → 40
tank/<device-id>/level/3      → 90
tank/<device-id>/pump         → ON | OFF  (state)
tank/<device-id>/valve/2      → OPEN | CLOSED
tank/<device-id>/cmd          ← {"pump":"ON"} | {"valve":2,"open":true} | {"auto":true}
tank/<device-id>/config       ← {"tanks":3,"low":20,"high":95,"maxRunMin":30}
tank/<device-id>/alert        → "Tank 2 sensor error"
```

Firmware tihtur

- WiFiManager — customer-in a WiFi a phone atangin a dah theih (code thlak ngai lo).
- Tank zat, level threshold, max time chu **config**-ah — in hrang hranga hralh nan code thar ngai lo.
- Sensor reading 5 average (median) — tui fawn avanga reading buai laka himna.
- Internet a thih pawhin auto mode a kal zel (ESP32-ah logic awm).
- OTA update — box hawng lovin firmware thar dah theih.

10. Scalability leh hma lam

- **Tank 4–6:** ESP32 pin a la tawk, relay 8CH-ah thlak mai.
- **Tank tam zawk:** GPIO 21/22-ah I2C expander (PCF8574).
- **Wire awm lo:** nakin sensor node wireless (ESP32-C3 + battery + solar, ESP-NOW) belh theih; main box code a inang.



Wireless sensor node (hma lam option, v2)

11. Siam dan kalphung (roadmap)

Step	Hna	Zawh chian
1	Component lei (BOM)	Thil zawng zawng a thleng
2	Bench test: ESP32 + sensor 1 + relay	Level % serial-ah a lang, relay a click
3	Buck 5.0V siam, power test	Multimeter-ah 5.0V, a lum lo
4	Valve 1 + relay test (pump awm lovin)	App/serial atangin a hawng/khar
5	Auto logic + safety rules code	Bench-ah flowchart ang takin a kal
6	MQTT + app (Blynk emaw PWA)	Phone atangin level en, control theih
7	Box-ah dah, CAT6 wiring	Label, gland, terminal a fel
8	Electrician-in pump/contactor zawm	MCB, earth a dik
9	Site test ni 7	Tank 3 auto-in a khat, alert a kal
10	Brochure, man bituk, customer hmasa	Install pakhat a zo

12. Image prompt-te (ChatGPT)

Brochure image siam leh nan. Image dik tawh pahnih hi document-ah a awm tawh (cover leh page 8).

Prompt — installation view

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Photorealistic image of a smart water tank automation system on the flat concrete rooftop of a
house in Mizoram, hills in background, daytime. Three plain black 1000 L water tanks in a row
(no brand names). A black waterproof ultrasonic sensor on each tank lid with a small white IP65
junction box. A pump at ground level with a suction pipe into a sump. One blue PVC main pipe runs
along the wall with three vertical branches, one beside each tank, rising to a 12V motorized
ball valve at each tank inlet. Grey CAT6 cables and red 12V wires run neatly down each tank to a
grey IP65 "Main Controller" box on the wall, with an MCB beside it and ONE black power cable to
the pump. No pipe touches the electrical box. A hand holds a phone app: Tank 1 75%, Tank 2 40%,
Tank 3 90%, Pump ON, Valve 2 ON, Valves 1 and 3 OFF, Auto mode ON. Legend panel: 3 x Ultrasonic
Sensors (JSN-SR04T), 3 x Motorized Ball Valves (12V), 1 x Water Pump, 1 x Main Controller (IP65),
3 x CAT6 Cables, 3 x 12V Control Wires, 1 x Power Cable. Product marketing photo style.
```

Prompt — main box close-up

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Premium product photography. Open grey IP65 enclosure with transparent lid labelled "Main
Controller". Inside: DC jack terminal (top left), blue LM2596 buck converter showing "5.0",
ESP32 DevKit with blue LED, blue 4-channel relay (SRD-05VDC, LEDs 2 and 4 lit), small perfboard
with six resistors, MCB + 25A contactor on DIN rail, labelled terminal row: 12V, GND, VALVE 1,
VALVE 2, VALVE 3, PUMP, TANK 1, TANK 2, TANK 3. Three CAT6 cables, red valve wires, one pump cable
and one mains cable through bottom cable glands. Tidy cable ties. Avoid breadboards, Arduino,
batteries, misspelled labels, water pipes.
```

AI image hi brochure/customer hmuhtir nan chauh. Wire zawm tak tak nan chuan he document-a diagram leh pin table hmang rawh.